HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323 | Course Title: Numerical Analysis
Mid-Term Examination | Semester: Fall 2024
Date: 4/1/2025 | Total Time: 1.5 Hours. | **Total Marks: 30**

Answer any three (03) of the following questions: $3 \times 10 = 30$

1. (a) Explain the operators E, ∇ and Δ. [3]
   (b) Find the approximate root of the equation $f(x) = x^3 - x - 1 = 0$ by using bisection method. $[a = 1, b = 2]$ [3.5]
   (c) Prove the given relations [3.5]
   (i) $(1 - \nabla)(1 + \Delta) = 1$.
   (ii) $E\nabla = \nabla E = \Delta$.
2. (a) Write down a short note of the false position method. [2]
   (b) Find the approximate root of equation $f(x) = x^3 - 5x + 3 = 0$ by using the false-position method. [4]
   (c) Find the approximate root of the equation $f(x) = x^3 - 6x + 4 = 0$ by using the Newton-Raphson method. [4]
3. (a) State the Lagrange's interpolation formula for $n + 1$ data. [1]
   (b) Find the value of the function $y = f(x)$ for $x = \alpha = 12.35$ and x for $f(x) = \sqrt{155}$ from the following table [using Lagrange's interpolation formula]: [9]

x	150	152	154	156
$y = f(x)$	12.247	12.329	12.410	12.490

4. (a) Find the number of students who obtained less than 45 marks from the following table: [4]

Marks (x)	30-40	40-50	50-60	60-70	70-80
Population $f(x)$	31	42	51	35	31

   (b) What is interpolation? Find the value of the function $y = f(x)$ for $x = 12$ from the following table: [6]

x	5	11	27	34	42
$f(x)$	23	899	17315	35606	68510

5. (a) Write the procedure of the Gauss-Seidel method to solve a system of equations [3]
   (b) Solve the following system of equations by using the Jacobi and Gauss-Seidel method: [7]

$$10x - 5y - 2z = 3$$
$$4x - 10y + 3z = -3$$
$$x + 6y + 10z = -3$$

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE325: System Analysis and Design

Batch: 22^{nd} | Semester: Fall 2024
Time: 1 Hour 30 Minutes
Date: 01/01/2025 | **Mid-Term Examination** | Total Marks: 30

Answer any five questions of the followings. $6 \times 5 = 30$

1. (a) Explain the system designers' view of information system. 3
   (b) Describe the skills needed by a system analyst. 3

2. (a) What is meant by Project Initiation? Write the questions that need to be asked regarding project selection. 3
   (b) List some business questions that can help a system analyst to identify the business objectives of a system. 3

3. (a) Why feasibility analysis is required to initiate a project? 2
   (b) Illustrate the steps in choosing hardware and software. 4

4. (a) Write the advantages and drawbacks of Bring Your Own Device (BYOD) option in system development. 3
   (b) List the situation when each of the following analysis can be applied: 3
      i. break-even analysis
      ii. cash-flow analysis
      iii. present value analysis

5. (a) Explain different types of Benefits and Costs of system. 3
   (b) Without considering present value, the benefits and the costs as follows: 3

Year	1	2	3	4	5	6
Costs	41000	44000	46000	52000	56000	61200
Benefits	25000	32500	39500	53000	71000	82000

Calculate the present value if the discount rate is 0.13.

6. Zihad owns a frozen-foods company and wants to develop an information system for tracking shipments to warehouses. The information of the development process is given below:

Task	Must Follow	Time (Weeks)	Crash Time	Cost Per Week
A	None	5	3	\$500
B	A	3	2	\$300
C	None	2	2	\$400
D	B, C	3	2	\$750
E	B, C	4	4	\$200
F	E	4	3	\$250
G	D	5	4	\$400
H	F, G	3	2	\$380

(a) Draw the PERT diagram for the system and identify the critical path. 3

(b) If Zihad expedites the project as much as he can, calculate the saving amount and the minimum time the project will take. 3

7. (a) List the techniques use to estimate the time it takes to complete each of the tasks of a system. 2

(b) Draw the fishbone diagram used to identify all the things that can go wrong in developing a system. 4

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 323: Operating System

Batch: 22nd

Semester: Fall 2024

Time: 1 Hour 30 Minutes

Date: 08/01/2025

Mid-Term Examination

Total Marks: 30

Answer any **five** questions of the following. **$5 \times 6 = 30$**

1. (a) Explain the role of Operating System (OS) as **"an extended machine"** and **"a resource manager"**. 3

   (b) Mention the role of OS in computer system structure with proper figure. Briefly explain about batch operating system. 3

2. (a) What is Kernel? Explain dual mode operation with figure. 3

   (b) Explain about Monolithic Kernel and Microkernel with diagram. 3

4. (a) Write the criteria of CPU Scheduling. 2

   (b) Processes are given with arrival and burst time on the following table. The time quantum of the system is 5 **units**. Apply Preemptive Round Robin Scheduling Algorithm and answer the following questions. 4

Process ID	Arrival Time	Burst Time
P1	0	7
P2	1	4
P3	2	15
P4	3	11
P5	4	20
P6	4	9

   i. Draw the Gantt chart with Ready Queue.
   ii. Calculate average Waiting time.
   iii. Calculate average Turn Around time.
   iv. Calculate Throughput in percentage.

5. (a) What is process? Explain Process State Transition with figures. 3

   (b) What is PCB? How does CPU switch from Process to Process? 3

6. (a) What is system calls? Explain system call execution. 3

(b) From the concept of fork() system call, determine how many times the printf functions will be executed in the following program. Explain your answer using a tree hierarchy. 3

```
#include <stdio.h>

int main() {
        fork();
        fork();
        printf("child process one");
        fork();
        printf("Child process two");

        return 0;
}
```

7. In the following table, there are 7 processes with their arrival time and burst time. The time quantum of the system is **2 units**. 6

Process ID	Priority	Arrival Time	Burst Time
1	2	0	3
2	6	2	5
3	3	1	4
4	5	4	2
5	7	6	9
6	4	5	4
7	10	7	10

Now calculate waiting time by using both FCFS and Priority Scheduling Algorithms (non preemptive) and determine which algorithm is better in terms of faster process execution.

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science Engineering and Technology
Department of Computer Science and Engineering(CSE)
MGT 417: Industrial and Operational Management
Mid-Term Examination
Fall 2024

Time: 1 Hour 30 Minutes

Total Marks: 30

Instructions:

- Answer any 3 sets of questions out of the 4.
- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.

No.	Question/ Statement	Marks
1.	a) Define the following terms- i. Industry ii. Management iii. Operation management	3
	b) Describe the similarities and dissimilarities of manufacturing and servicing.	4
	c) Describe the operation management trends sustained in different industries in Bangladesh.	3
2.	a) Define productivity.	2
	b) Discuss a production system with a block diagram.	4
	c) Discuss the classifications of the production system.	4
3.	a) What is plant layout? Discuss the different types of plant layouts.	4
	b) Explain the life cycle of a product.	3
	b) How an organization can gain a sustainable competitive advantage through its operations.	3

4.	a) Show the mathematical relationship between profit and productivity. As well as discuss how breakeven, loss, and profit impact productivity through the equation.	**5**
	b) Mathematically show that which company is more productive. <u>Scenario- 1</u> Company A sells a table of 75000 tk which costs them 65000 tk. <u>Scenario- 2</u> Company B produces 1500 tablets which took them 1250 hours.	**5**

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 325: Computer Peripherals & Interfacing

Batch: 22nd | Semester: Fall 2024

Mid Term Examination

Time: 1 Hour 30 Minutes

Date: 13/01/2025

Total Marks: 30

Answer any **five** questions of the followings: $5 \times 6 = 30$

Marks

1.	(a)	Define computer peripherals with examples.	2
	(b)	Why peripheral devices are significant for computation?	2
	(c)	Wirte short notes on **(i)** Scanner and **(ii)** Digitizer.	2×1=2
2.	(a)	Mention the key features of: **(i)** Drum plotter **(ii)** Flat bed plotter and **(iii)** inkjet plotter.	3×1=3
	(b)	Differentiate between **Liquid crystal display** and **Cathode ray tube.**	3
3.	(a)	Write the jobs of peripheral adapter and give illustration.	3
	(b)	Mention the jobs of three key terms of data highways.	3
4.	(a)	Briefly write on **(i)** Point sensor **(ii)** Distributed sensor **(iii)** Reflective sensor.	3×1=3
	(b)	Write the pros and cons of optical sensor.	3
5.	(a)	What is the job of a Transducer.	2
	(b)	Mention the key differences between bit rate and baud rate, along with their measuring formulas.	4
6.	(a)	Explain how an asynchronous transmission process is executed?	3
	(b)	Write the wroking principle of Stepper Motor Circuit.	3
7.	(a)	Why graphics system is necessary?	2
	(b)	Explain the Raster scan display process with proper illustration.	4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE321:Theory of Computing

Batch: 22nd | Semester: Fall 2024

Time:1 Hour 30 minutes

Date: 06/01/2025 **Mid-Term Examination** Total Marks: 30

Answer any **five** questions. Marks: 5×6 = 30

Marks

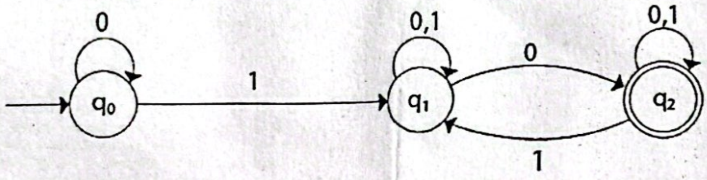
1. (a) Write the uses of automata. 2
   (b) State the important issues that define automata. 4

2. (a) Compare between $\sum^{+}$and $\sum^{*}$, when $\sum$={abc}. 2
   (b) Draw an NFA for a string that starts and ends with the same symbol, over the alphabet $\sum$ = {a,b}. 4

3. (a) With proper diagram, differentiate between NFA and DFA. 3
   (b) Write short notes on 3×1=3
      i) Alphabet
      ii) Power of Alphabet
      iii) Concatenation of string

4. Design DFA for the following strings: 2×3=6
   a) Starts with **'abc'** , over the alphabet $\sum$ = {a,b,c}.
   b) Ends with **'000'** , over the alphabet $\sum$ = {0,1}.

5. *Convert the following NFAto a DFA:* 6



6. Design a DFA accepting the set of all strings with **'abc'** substring, over the alphabet {a,b,c}. Show that **bbcacabc** is accepted by the system. 6

7. Using DFA minimization techniques, minimize state(s) of the given DFA. 6